recombination and thermal ionization create feedback loops acting at different time scales through which the CB-electron density, ρ_c, and T_{equil} act back on the energy deposition process [**Fig. 1(b)**]. The respective feedback strength is given by the dependence of avalanche-ionization and recombination rates on free electron density, and by the temperature dependence of thermal ionization. In the following,

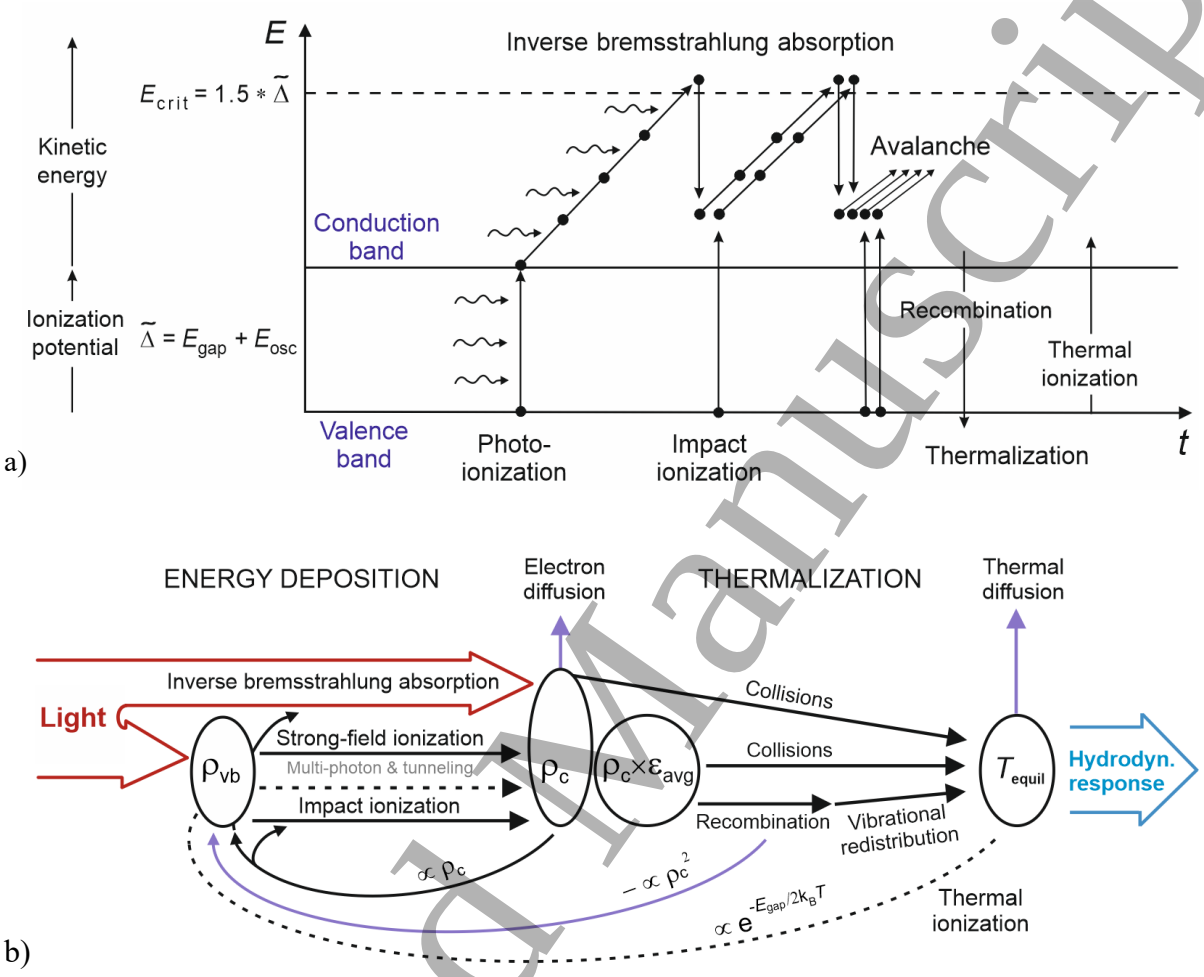

Figure 1 Pathways of nonlinear energy deposition and thermalization during optical breakdown in large-bandgap dielectrics. (a) Visualization of the pathways within the band structure of the dielectric, (b) Visualization with feedback loops between the individual mechanisms. The energy flow from the laser light into the electronic system of the dielectric is followed by thermalization and the hydrodynamic response of the medium. These pathways include feedback loops acting at different time scales through which the free-electron density ρ_c and temperature T_{equil} produced by nonlinear absorption and thermalization act back on energy deposition and ionization. When the thermodynamic temperature of the medium is sufficiently high, nonlinear absorption is complemented by thermal ionization. Pathways illustrating the coupling of incident laser light into VB and CB electrons are marked in red, and pathways indicating CB electron losses and heat dissipation are marked in violet. While the feedback of free-electron density on the avalanche ionization rate occurs on a fs time scale, the recombination time ranges from < 1ps to several ps (depending on ρ_c), and the thermalization time as a whole amounts to ≈20 ps. Recombination, thermalization, and thermal ionization play no role during fs breakdown but become increasingly important for longer pulse durations. The hydrodynamic response (shock wave emission and cavitation) sets in when the energy of the free electrons is thermalized.

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we will see how the different time constants of primary energy deposition processes and feedback loops result in different features of the breakdown dynamics depending on laser pulse duration and wavelength.

Strong-field ionization provides seed electrons quasi-instantaneously. Seed electron generation by MPI dominates from UV to near-IR and at low irradiance values, whereas tunneling becomes relevant at longer wavelengths [6,22,145-147]. The dependence of strong-field ionization on irradiance, I, becomes complicated, when tunneling gets involved but when multiphoton ionization dominates, it is simply proportional to I^k, where k is the order of the multiphoton process [113,144]. With increasing wavelength, simultaneous absorption of more and more photons is required to cross the band gap, and the multiphoton ionization rate η_{MPI} decreases.

For bubble formation at $\tau_L \geq 100$ fs, most electrons are produced by *avalanche ionization* [37,143]. This process is subject to time constraints because impact ionization can occur only after conduction band (CB) electrons have gained sufficient energy through inverse bremsstrahlung absorption. Free electrons can absorb photons only during collisions with heavy particles, which occur in intervals of ≈ 1 fs [37,113,115,148]. Since several subsequent absorption processes are needed to collect sufficient energy for impact ionization, each doubling sequence of free electron density takes several femtoseconds and the feedback of free-electron density on the avalanche ionization rate has a time-lag [113,146]. Nevertheless, since it occurs on a fs time scale, this feedback loop plays a significant role already in fs breakdown. The energy distribution of CB electrons is initially nonstationary and its evolution during this time has to be described by a multi-rate equation [143,146,149]. However, for sufficiently long pulse durations the energy distribution evolves asymptotically into a stationary distribution, whereby the transition time depends on bandgap, wavelength, and irradiance. For water, this time amounts to about 100 fs [37,149]. Once the stationary regime has been reached, avalanche ionization is proportional to the CB electron density and approximately proportional to I [37,113,150]: $(d\rho_c/dt) = \eta_{AI} \times \rho_c\, I$. The wavelength dependence of AI is contained in the avalanche ionization rate, η_{AI}. With increasing wavelength, the photon energy decreases and more and more inverse bremsstrahlung events are needed to provide the energy required for impact ionization but the rate constant for individual IBA events is approximately proportional to λ_L^2 [37,38,113]. Both relations together yield a relatively weak overall dependence of the AI rate η_{AI} on wavelength [37].

The pulse duration dependence of plasma-mediated ablation and bubble formation is determined by an interplay of SFI and AI. Time constraints of AI play a significant role for the pulse duration dependence of optical breakdown thresholds in the fs range. For pulses between 100 fs and 5fs, Lenzner

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et al. reported a relatively slow drop of the radiant exposure ablation threshold F_{th} with decreasing τ_L, corresponding to a fast increase of the breakdown irradiance I_{th} [126]. This behavior reflects the growing time constraints of AI with decreasing τ_L, which goes along with an increase of the relative importance of SFI. However, Lenzner et al. did not take the time-lag of AI into account and attributed the resulting mismatch between theoretical predictions and experimental observations to deficiencies of the Keldysh model for SFI.

Time constraints of AI were first discussed by Kennedy [113] and are included in all models that explicitly treat the gain of kinetic CB electron energy as a succession of IBA events separated by a finite collision time. Coupling of laser energy into CB electrons during collisions and collisional losses are key elements in the description of AI [6,125,142,151-153]. Stuart et al. showed that an approximate description of these processes by a rate equation based on the Drude model yielded good results with less numerical expense than full models based on solving kinetic equations [125]. Noack et al. adapted the rate equation approach to breakdown in water and considered the finite time needed to reach the impact ionization energy [115]. Simulations considering SFI initiation, AI, and recombination led to good agreement with experimental data for pulse durations between 100 fs and 76 ns. Vogel et al. then combined numerical simulations of plasma formation in water with a model of thermoeleastic bubble formation [22,117]. Rethfeld introduced a multirate equation (MRE) model, which describes the temporal evolution of the CB electron energy distribution and ionization avalanche rate during the laser pulse. As already mentioned, the MRE approach is particularly important for short pulse durations below 100 fs and at low irradiance [146,153]. Christensen & Balling modified the approach to make it fully consistent with the law of energy conservation [143], and Liang et al. adapted the MRE model to breakdown in water considering the complex band structure of water [149], which contains an intermediate energy level for solvated electrons that facilitates breakdown initiation. Linz et al. modeled the wavelength dependence of ns and fs breakdown in water taking this band structure into account [36,37]. Model fits to experimental $I_{th}(\lambda)$ data for IR ns breakdown provided evidence that AI is initiated by multiphoton ionization rather than through background electrons or impurities. Fits to experimental results for fs breakdown at 250 fs pulse duration showed a dominant role of AI over SFI in the entire investigated wavelength range from 335 nm to 1085 nm.

The time scale for *recombination* is longer than the time constants relevant for avalanche ionization. In liquid water, recombination proceeds mainly as non-radiative interaction of CB electrons with $H_3O^+_{aq}$ ions and as electron attachment to neutral OH fragments [37,154,155], while in plasmas with

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